CODESCHOOL · COMPARATIVE RESEARCH

US vs CBSE: CT and AI

How the USA teaches computational thinking and AI, from which grade, how it differs from CBSE's Classes 3-8 curriculum, and what India and CodeSchool can do next.

Deep research report · 6 November 2026

Built on 5 research threads plus the CBSE verification and handbook review

Clickable contents, bookmarks and live source links

Repository: github.com/evolvian2026/CodeSchool

Summary

India mandates what America merely recommends

PK/K
US model standards start unplugged CT at age 4-5 (CSTA 2026, **voluntary**)

Class 3
CBSE starts CT for a whole board in 2026-27 (50 h a year in Classes 3-5)

Grade 5 vs Cl. 8
US children train an ML model by Grade 5; CBSE's first hands-on model training is Class 8

12 of 50
US states require a CS course to graduate; none requires elementary AI

0 of 12
CBSE CT&AI books with any coding; first programming subject is Class 11

60% / 6.1%
US high schools offering CS vs students enrolled (2024-25)

The United States writes computational thinking (CT) and AI into its model standards from pre-kindergarten, but almost nothing requires a school to teach them. The 2026 CSTA PK-12 standards start unplugged algorithms at age four or five, block coding in Grade 1, loops in Grade 2, conditionals in Grade 3, and have children train a machine-learning model themselves by Grade 5. Yet the standards are voluntary. Only 12 states require even one high-school CS course to graduate, and only 4 states write AI into their CS standards. Federal action is limited to executive orders, competitions and grant preferences. Delivery runs on free platforms from non-profits and companies, one-day teacher workshops and near-universal school devices. The result is wide availability, thin take-up (60% of high schools offer CS, 6.1% of students enrol) and no national measure of what children learn. CBSE has inverted this model. Its CT&AI curriculum for Classes 3-8 (2026-27) is effectively compulsory for a whole board and gives more hours than US standards assume: 50 a year in Classes 3-5 and 100 in Classes 6-8, against 20-40 for US elementary grades. But its free handbooks contain **no coding, no project materials and no rubrics**, hands-on AI appears only in Class 8, and a CBSE student meets no programming language in any subject before Class 11. India's wider system adds a device gap (64.7% of schools have computers), a cascade-training model for more than a crore (10 million) teachers, no assessment of CT, programmes split across ministries, and state pilots in Kerala and Tamil Nadu that already teach more coding than CBSE does. The improvements that matter are a single policy pathway, a "build it" layer in the middle years, coaching rather than one-day training, funding for working labs rather than sanctioned ones, and a cheap CT diagnostic. For CodeSchool, the comparison gives a clear brief. Sell the layer CBSE left blank (coding tied to each puzzle chapter, project kits, AI teacher support, hands-on AI that works with few devices, and assessment), use the US grade progression as the design reference for levels, and treat evidence of learning, not content, as what the school pays for.

The one-line finding

Mirror images. The US built content, tools, teacher networks and devices but obliges few children to use them, and has no national test. CBSE set an obligation and generous hours for a whole board, but supplied only puzzle banks to fill them. Each system's missing piece is the other's strength.

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1. The US standard starts at age four, but no one has to follow it

A year-by-year spine with AI from Grade 1

The current US reference document is the **2026 CSTA PK-12 Computer Science Standards**, published in summer 2026 and launched at CSTA's July conference. About 25 writers from 18 states drafted it, with roughly 450 reviewers (2026 CSTA PK-12 Standards). It replaces the 2017 standards, which grouped grades into multi-year levels (K-2, 3-5, 6-8, 9-10 and 11-12) and contained almost no AI. Their only two AI standards sat in Grades 11-12 (CSTA 2017 Standards). The 2026 version has a PK/K band, separate standards for each of Grades 1-5, a single middle-school band, a foundational high-school band, and high-school specialty tiers in six areas, one of which is Artificial Intelligence. It has about 194 foundational standards under five concepts: Algorithms & Design, Programming, Data & Analysis, Systems & Security, and Computing & Society. It also has twelve practices, four of them labelled computational thinking: define problems, abstract, create, and test and refine. The writers deliberately **wove AI through every concept instead of making it a separate topic**. The document also says plainly that CS "is not" using word processors, slide decks or generative-AI tools, and that CS "is what powers AI" (2026 CSTA Standards).

The grade placements matter most for a CBSE comparison. US Grade *n* and Indian Class *n* cover roughly the same ages (CSTA puts Grade 3 at ages 8-9), so the table can be read row by row.

Element	US start (CSTA 2026, voluntary)	Example standard	2017 placement
Unplugged algorithms	PK/K	EK-ALG-PS-01 "Carry out algorithms in daily activities"; computers "not required"	K-2
Block coding (sequence)	Grade 1	E1-PRO-PD-04: translate a sequence "into block-based code"	K-2
Loops	Grade 2	E2-PRO-PD-04: "sequence, events, and iteration"	K-2
Conditionals in code	Grade 3	E3-PRO-PD-04: code with "selection" ("conditionals [if/then statements]")	3-5
Variables used in programs	Grade 5	E5-PRO-VD-06 "Use variables to store, compare, and modify data"	3-5
Procedures	Middle school	MS-PRO-PD-12, without parameters	6-8, with parameters
Text-based programming	Optional from about Grade 5; never required	E5 boundary: "block-based or beginner text-based environments"	Never mandated
AI recognition	Grade 1	E1-ALG-ML-02: AI "use[s] patterns in data to make decisions"	Grades 11-12 only
How training data shapes a model	Grade 2, with bias informally by Grades 3-4	E2-ALG-ML-02: "limited or unbalanced examples can lead to unfair or uneven results"	None
Child trains a model	Grade 5	E5-ALG-ML-02 "Train a machine learning model to make a classification or prediction"	None
Using AI tools; checking AI-written code	Middle school	MS-ALG-PS-05; MS-PRO-RD-18 "Analyze AI-generated code for accuracy"	None

Source for all rows: 2026 CSTA Standards and 2017 Standards.

Two features stand out. First, **the machine-learning strand advances a step every year from Grade 1 to Grade 5**. It moves from "AI uses patterns" to how data trains a model, what happens when data is added, the term "bias", and finally training a classifier and asking "who is missing from the training data". Second, the standards **never require text-based code** at any foundational grade. The 2026 revision actually moved some programming later than 2017: loops from K-2 to Grade 2, parameters from middle to high school. Programming also shifted toward "reading, evaluating, modifying, and debugging programs" rather than only writing them (2026 CSTA Standards). The standards assume **20-40 hours of CS a year in PK-5**, and the equivalent of a year-long course in middle school.

A deeper, more technical layer sits beside CSTA. The AAAI/CSTA **AI4K12 "Five Big Ideas"** charts (perception, representation and reasoning, learning, natural interaction, societal impact) put training a classifier and checking its accuracy in Grades 3-5. In Grades 6-8 they add clustering, decision trees versus neural networks, and 1-3-neuron networks. The official site still calls all five charts drafts open for comment (AI4K12; Big Idea 3 chart). Other frameworks treat CT as a general practice rather than a grade sequence. NGSS makes "Using Mathematics and Computational Thinking" a science practice from K-12 (NGSS Appendix F). ISTE's student standard 1.5, "Computational Thinker", has no grade bands at all (ISTE). Internationally, the EC/OECD **AILit framework** (4 domains, 19 competences, finalised in 2026 with support from CodeAI, the new name for Code.org) will inform PISA's first assessment of media and AI literacy, in 2029 (AILit; EdTech Innovation Hub).

Policy and money work through incentives, pledges and one-course mandates

None of this binds a school. CSTA says its standards "do not prescribe specific curriculum, instructional materials, or assessments" (2026 CSTA Standards). States decide, and states mostly encourage rather than require. CSTA counts **46 states with CS standards in 2026, against 7 in 2017**, though other counts range from 42 to 44 depending on date and definition. Code.org's latest State of AI + CS Education report (data finalised December 2025) found **12 states requiring a CS course to graduate** and none requiring a combined AI-and-CS course. Only **4 states (Colorado, Virginia, North Dakota and Ohio) emphasise AI in their CS standards**, 17 states have guidance calling CS fundamental to AI literacy, and 4-5 states fund AI+CS teacher training (K-12 Dive; EdWeek). In 2024, 32 states required every high school to *offer* CS, and state budgets set aside more than $88 million for CS (K-12 Dive, 2024). Below high school, mandates are rare and are usually requirements to offer CS, not sequences every child must follow. The 2024 report counted 17 states that had extended course requirements to middle or elementary schools (K-12 Dive, 2024). Some states, such as California, leave elementary CS entirely to each school (Frontiers in Education).

AI law is growing fast but mostly governs how AI is *used*, not what is *taught*. About **35 states plus Puerto Rico** had official K-12 AI guidance by August 2026, nearly all of it voluntary (AI for Education tracker). In 2026, legislators filed 87 AI-in-classroom bills in 28 states and enacted 13 of them. Only two clearly require instruction: Alabama HB 329, a CS course with AI to graduate from the class of 2032, and Utah HB 218, a required Grade 7 or 8 digital-skills course with AI literacy. Ohio, Oklahoma and Maryland instead require district AI policies or coordinators (FutureEd; Ohio DEW). **No US state law found requires AI teaching in elementary grades**, so CBSE's Class 3 start is ahead of every US mandate on record.

Federal levers are incentives. Executive Order 14277 (23 April 2025) created a White House Task Force on AI Education and a Presidential AI Challenge (Federal Register). The first Challenge drew more than 20,000 students (WhiteHouse.gov), and it was relaunched on 13 October 2026 as the "Presidential Super Intelligence Challenge" with a new school readiness seal (MeriTalk). The Department of Education told states that existing grant money may be spent on AI, and finalised an AI "supplemental priority" for discretionary grants on 13 April 2026 (Federal Register, 91 FR 18774). That is a scoring preference in grant competitions, not a curriculum rule (AEI). Most of the money comes from companies. In September 2025, about 51 organisations made self-reported pledges: Google $1 billion for education and job training (including $150 million in grants), NVIDIA $25 million over five years, Microsoft a free year of Copilot for K-12, and Anthropic $1 million over three years, among others (WhiteHouse.gov). No independent audit of whether the pledges were kept was found. Obama's 2016 "CS for All" asked Congress for $4 billion, which Congress never enacted as proposed (Obama White House archive). The structural point is that **America's CS curriculum is supplied by non-profits and companies, and the state's role is to permit and encourage it**.

2. Free platforms, one-day workshops and a device for every child

US classroom practice runs on free curricula, chiefly Code.org's (now renamed CodeAI). **CS Fundamentals** has six courses (A-F) covering K-5, each with 13-17 lessons of about 45 minutes. Course A covers loops and events for K-1, Course D brings conditionals in Grades 3-4, and Course F brings variables in Grades 4-5. Lessons are mostly block-based puzzles on devices with some unplugged discussion, and a separate fully unplugged track exists as a supplement (Code.org CSF). Code.org recommends about **20 classroom hours a year** (Code.org PL guide). In middle school, CS Discoveries has become **AI Discoveries**. It can run for a quarter, a semester or a full year, with units on thinking critically about AI, machine learning, web development, games, data and physical computing. It includes an AI tutor that gives hints without giving answers, and an AI assistant for teachers (Code.org AI Discoveries). Hour of Code became **Hour of AI** in December 2025. CSforALL reported 15.5-17.9 million participants, two inconsistent figures, both self-reported (CSforALL).

Around Code.org sits a set of short, free AI modules. **MIT's Day of AI** offers 4-8 hours per grade band and already has an India page (Day of AI impact report; Day of AI India). **Experience AI**, from the UK's Raspberry Pi Foundation and Google DeepMind, provides 4-6 lessons for ages 11-14 and publishes Indian-English units (Experience AI). Common Sense publishes 20-minute AI lessons, and Charlotte-Mecklenburg requires every Grade 3-5 pupil to complete "What is AI?" (Common Sense; CMS). Full AI electives exist but are rarer. Georgia's NSF-funded AI4GA is a nine-week middle-school course that has reached about 1,600 students (NSF PAR). Free platforms can also disappear: **Google retired CS First on 30 June 2025 and deleted its data**, sending teachers to the Raspberry Pi Foundation's Experience CS (Google CS First Help).

Schools deliver this in different ways. Some use weekly "specials" rotations of 50-150 minutes, as in Gwinnett County, Georgia, where every K-5 pupil at award-winning schools gets CS and STEM specials. Others fit it into the classroom teacher's day, and Arkansas embeds K-8 CS across subjects (Hanover Research; GCPS; Arkansas DESE). No national figure for weekly elementary CS minutes exists. Whichever model a school uses, it relies on hardware: about **88% of US public schools run 1:1 device programmes** (NCES School Pulse Panel, via a secondary repost) (Education Research Report). The US can assume children are working on devices. India cannot.

US teachers are mostly non-specialists. In 2020 only **82 of more than 150,000 newly certified US teachers received a CS certification** (AIR). Only about 30% of CS teachers held a computing degree (CSTA/Kapor), and in 2025 half had no CS colleague at their school (CS Teacher Landscape). Code.org's answer is "no specialists are needed": a free curriculum plus one-day Regional Partner workshops, attended by more than 100,000 CS Fundamentals teachers (Code.org CSF). AI training is catching up unevenly. By fall 2025, 50% of teachers had attended at least one AI training session (EdWeek). Gallup-Walton found 68% got no school-provided AI training in 2024-25 (Walton Family Foundation). RAND found teacher AI training in 67% of low-poverty districts against 39% of high-poverty ones, almost always optional (AI for Education summary of RAND). Big-tech academies are scaling up. The AFT teachers' union academy has $23 million from Microsoft, OpenAI and Anthropic and aims to train 400,000 educators (Chalkbeat). Google and ISTE+ASCD offer free AI training to all 6 million US educators (Google blog). The US model therefore rests on free curriculum, short training sessions and devices. It scales reach, but it leaves quality to chance.

3. Sixty percent access, six percent uptake and almost no measurement

The US data shows that availability does not mean participation. **60% of public high schools offered foundational CS in 2024-25**, unchanged from the year before, yet only **6.1% of students enrolled**. Young women were 34% of enrolees (K-12 Dive). A secondary analysis of the same data shows the gap by school size: 91% of large high schools offer CS against 44% of small ones, which are disproportionately rural (aiadoption.org). Participation rose only where states mandated it. In graduation-requirement states, participation climbed from 9% to 10.4% of eligible students (EdWeek). Lower grades are counted less reliably. In 2024, 8.3% of middle-school and 13% of elementary pupils took CS in a year, an undercount because CS integrated into other subjects is excluded (CSTA).

Assessment is the weakest part of the US system. NAEP's Technology and Engineering Literacy test, the only national measure (Grade 8, given in 2014 and 2018), was **voted to be discontinued in 2023** because its update cycle could not keep pace with technology (NAGB). No state was found that tests K-8 CS. In classrooms, assessment means lesson checks and graded projects inside the curriculum; Code.org's Course B, for example, ends with a game each student builds (Code.org CSF). The only outcome data at scale is self-selected. In May 2025, **175,935 students took AP CS Principles (61.7% scored 3 or above) and 93,906 took AP CS A (67.0%)** (College Board CSP; College Board CSA). From 2027-28, AP CSP adds an "AI Create Project" in which students build an app that uses AI (GovTech). The closest US match to CBSE's puzzle style is Bebras, an optional November contest (in a 2020 format, 15 multiple-choice CT questions in 45 minutes) (Bebras USA). In the US, puzzles are enrichment; in CBSE they are the core.

The causal evidence carries a warning for India. A Maryland study found that taking a high-school CS course raised the chance of a CS bachelor's degree by 5 percentage points (EdWorkingPapers 24-904); advocates restate this as at least an 8% rise in earnings at age 24 (PR Newswire). A May 2026 Arkansas working paper reportedly finds that the state's 2015 rule that every high school offer CS raised course-taking but had **no detectable effect on college, choice of major or earnings** five years on. That finding is provisional, because only fragments of the paper could be read (Arkansas DESE). Read together, the two studies suggest that **universal provision without quality does not reproduce the gains of good courses**, which bears directly on a board-wide mandate like CBSE's. At younger ages, meta-analyses report large effects of unplugged CT activities on CT test scores ($g = 1.028$ across 13 studies; $g = 0.631$ for K-9). Those studies are small and mostly use tests their own authors designed (Chen et al. 2023; ResearchGate 2025). An umbrella review of 33 syntheses found larger effects from project-based work and from programming itself (PMC umbrella review). No rigorous evidence exists, anywhere, on what AI-literacy lessons achieve.

Meanwhile children's use of AI has run ahead of any teaching about it. **54% of US teens have used chatbots for schoolwork** (Pew). RAND found 54% of middle and high schoolers using AI for school, but only 19% saying a teacher had explained how to use it, and only 45% of principals reporting an AI policy (RAND RR-A4180-1; EdTech Current). The US problem is the reverse of India's: tools are everywhere, but structured teaching about them is thin.

4. CBSE assigns more hours than America but supplies less to fill them

CBSE's curriculum, launched by Circular Acad-15/2026 on 1 April 2026, is "the prescribed curriculum". Schools are "advised" to implement it, and the books circular nine days later says it "mandates CT + AI + Ethics + Unplugged Learning" and that schools "should" use the free handbooks (Acad-15/2026; Acad-18/2026). It sets **50 hours a year in Classes 3-5**, inside Mathematics and The World Around Us (TWAU). In **Classes 6-8 it sets 100 hours a year**: 40 of CT, 20 of AI and 40 of projects. It is platform-agnostic and favours free tools (CBSE CT&AI curriculum). The 12 free handbooks, produced by IIT Gandhinagar's Centre for Creative Learning, are banks of maths puzzles, mostly multiple-choice, arranged chapter by chapter to follow the NCERT maths textbooks. Each class gets one or two unplugged activities adapted from CS Unplugged, such as a Caesar cipher in Class 3 and a weighing-scale search in Class 5 (CBSE CT&AI page; Class 3 Teacher Handbook). Classes 6-8 add four AI chapters each. The student books teach the concepts in text, while the teacher books contain only exercises and answer keys. Only Class 8 is hands-on, with three Teachable Machine labs (a waste classifier, voice commands and a cat/dog bias test) that need a computer, webcam and internet (Class 8 Student Handbook). CBSE's April 2026 webinar presents the absence of coding as deliberate. It promises "no 'write a program'" and "no expectation of computer access" for assessment, and for Class 6 says pupils "Recognise — not build" AI. It also lists an "AI Project list with rubrics" and AI "Concept Handouts" as coming soon; neither had been published by November 2026 (CBSE webinar slides).

Set against the US standards, the gaps are concrete and sit at the same ages.

Dimension	US (CSTA 2026 and common practice)	CBSE CT&AI, Classes 3-8 (2026-27)
Legal status	Voluntary model; states adopt or adapt	Board-wide curriculum: "advised" in Acad-15, "mandates" in Acad-18; no board exam
CT start	PK/K, unplugged	Class 3, maths puzzles
Hours	20-40 h/yr in PK-5; a year-long course per band in MS	**50 h/yr (3-5); 100 h/yr (6-8)**
Coding	Blocks from Grade 1; conditionals in code by Grade 3; variables by Grade 5	Scratch named as one Class 3-5 competency; **no coding in any of the 12 books**; webinar says "no coding/software"
If-then logic	Written in code in Grade 3	Puzzle rules from Class 3; formal "if–then reasoning" outcome in Class 7
First text language	Optional from about Grade 5; AP CSA in high school	**Class 11** (Python in CS 083 and IP 065); Classes 9-10 Computer Applications teaches office tools and HTML only
AI start	Grade 1 recognition; child trains a model in Grade 5	Class 6 concepts in text; first hands-on model training in **Class 8**
AI ethics and bias	Grades 2-4 informally; model cards in middle school	Class 6 digital ethics; Class 7 bias; Class 8 fairness and responsible AI
Projects	Built into curricula (games, apps)	40 h/yr in Classes 6-8; **no project materials in any book**; rubrics promised
Teachers	Non-specialists with one-day workshops and free scripted curricula	Maths and other subject teachers (3-5); subject teachers plus computer teacher (6-8); CBSE's own one-day CPD workshops (6 CPD hours)
Devices	About 88% of schools 1:1	64.7% of all Indian schools have computers; design assumes few devices
Assessment	Project and lesson checks; no state K-8 tests; NAEP TEL discontinued	Written tests, group activities, observation journal, practicals; **no templates or rubrics supplied**

Sources: 2026 CSTA Standards; CBSE CT&AI curriculum; CBSE webinar; CS 083; CA 165 Class IX; ThePrint on UDISE+.

CBSE's design deserves a fair hearing. It is **ahead of every US mandate** in coverage: one board requires CT from Class 3 for all its pupils, while no US state requires elementary AI and only 12 require one high-school CS course. It gives more time than CSTA assumes. Its unplugged, maths-embedded design is a rational answer to Indian infrastructure rather than a lack of ambition. UDISE+ 2024-25 puts **computers in 64.7% of schools and internet in 63.5%**. Bihar's government schools have computers in only 16.5% of cases, against 72% of its private schools (ThePrint), and more

than 25,000 schools lacked electricity (Education for All in India). The puzzles are also demanding. Class 3 includes finding the heavier of 12 balls in the fewest weighings, and Class 5 compares linear search with divide-and-conquer (Class 5 Student Handbook). The weakness is what happens next. Pupils only *follow* or *compare* algorithms; no book asks them to *write* one. The research favours hands-on programming and projects over unplugged work alone, and CBSE's 40 project hours a year come with nothing to fill them. The Class 9 CT&AI modules have already been dropped for the 2026-27 batch (Acad-70/2026). That leaves **five years, Classes 6-10, with no programming between the Classes 3-5 Scratch competency and Class 11 Python**, in exactly the years (US Grades 6-10) when US standards move pupils to procedures, data and critical use of AI.

India beyond CBSE: a national announcement, state pilots and a device gap

Nationally, NEP 2020 para 4.25 places "mathematics and computational thinking" throughout schooling and coding activities in Classes 6-8 (Rajya Sabha PQ). In October 2025 the Ministry of Education said "AI & CT" would be *introduced* from Class 3 in all schools from 2026-27. It did not use the word "mandatory", and state boards are not bound by CBSE (DD News; EducationWorld). The Ministry named NISHTHA as the training channel (LatestLY/IANS), but CBSE's own plan is one-day district and regional workshops, half-day lead-school sessions and videos on the DD PM e-Vidya channel (CBSE webinar). Several ministries run separate, voluntary AI programmes. MSDE runs SOAR (three 15-hour micro-credentials for Classes 6-12), MeitY runs YUVAi (selective, Classes 8-12), and MoE, CBSE and Intel offer the 4-hour AI for All course (PIB backgrounder; IndiaAI; Organiser). Some pieces are announced but not delivered: the 2025 Budget's 50,000 new Atal Tinkering Labs (just over 10,000 existed in December 2025) and NCERT's Class 11-12 AI textbooks (Business Today; MoE PIB; Tribune).

Some states teach more coding than CBSE. **Kerala's state ICT textbooks have Scratch games by Class 4, a Class 7 computer-vision chapter built in PictoBlox and Scratch, and Python in Classes 9-10** (ThePrint; Onmanorama). **Tamil Nadu's TN SPARK** grew from 85 pilot schools in September 2025 to about 5,000 government schools by July 2026. It uses bilingual Tamil-English textbooks with Blockly, Turtle Art and Scratch, taught by maths and science teachers (Business Today; NewKerala). Punjab, Telangana and Haryana have announced AI rollouts without published curricula. Karnataka offers AI as a vocational subject in Classes 9-10, and CISCE offers Robotics & AI as an elective in Classes 9-10 (Careers360; Sakshi Education; Careers360 on CISCE). Several of the 2026 state "AI" initiatives, such as Gujarat's data centre and Rajasthan's AI reading tests, *use* AI on pupils rather than teach it (Organiser).

The wider system has three constraints the US does not face at the same scale. The first is **teachers**. India crossed one crore teachers in UDISE+ 2024-25 (Vision IAS) and has 1,04,125 single-teacher schools (DoSEL/PIB). One IIT expert estimates that training even the first 5 million teachers needs about 10,000 master trainers and 6 lakh district or cluster trainers (The Secretariat). Where computer specialists exist, they are often contract staff; one Puducherry notice offered Rs 8,500 a month (Puducherry notice). The second is **working equipment**. An older Ministry review counted 1,20,614 ICT labs sanctioned but 78,498 functional, and 82,120 smart classrooms sanctioned but only 9,833 functional (DoSEL review minutes). At home, about 90% of rural 14-16-year-olds have a smartphone in the household, but only 27% of 14-year-olds own one, and boys are more likely than girls to have access (ThePrint on ASER 2024; Ideas for India). The third is **measurement**. India withdrew from PISA 2025 (The Wire), and no national survey was found that measures CT. Bebras India, run by ACM India's CSpathshala, is free, covers ages 8-19 and runs in 12 languages, but its national participation is in the tens of thousands; Goa alone ran it for more than 16,000 students through its SCERT (Bebras India; Herald Goa).

5. Five repairs would turn India's mandate into classroom capability

The comparison does not argue for copying the US. America shows what happens with excellent content and devices but no obligation: low uptake and no measurement. India has the obligation and the hours, but lacks the materials, the trained teachers and the feedback loop. The repairs target that missing middle.

Area	India's gap	Repair	Evidence behind it
Policy	"Advised" vs "mandates" wording; Class 9 deferred; MoE, MSDE, MeitY and NITI run separate programmes; state boards not bound	One published CT/AI pathway from Class 3 to 12 that maps SOAR, YUVAi, ATL and state pilots onto it; settle the Class 9-10 plan; publish the promised project rubrics and AI handouts	US participation rose only under mandates (EdWeek); Arkansas warns that mandates without quality add little (Arkansas DESE)
Curriculum	Pupils follow algorithms but never write one; AI hands-on only in Class 8; projects unsupported	Keep unplugged CT in Classes 3-5 but add a "build it" strand (blocks where devices allow, paper code where not); bring a hands-on data and ML activity into Classes 6-7; fill the 120 project hours	CSTA has children train a model by Grade 5 (CSTA 2026); project-based and programming interventions show larger effects (PMC umbrella review); Kerala and TN already teach Scratch in government schools
Teachers	One-day CPD; AI chapters lack teacher material; contract computer instructors	Scripted lessons inside the materials, a cluster-level coaching layer resourced through block and district training institutes, and CT/AI in pre-service teacher education	US one-day workshops scale reach but leave about half of teachers without AI training (EdWeek); IIT trainer estimate (The Secretariat); teachers' AI misconceptions persist (ORF)
Devices	About a third of labs and most smart classrooms sanctioned but not working; shared, gendered home phones	Release funds only for working labs (tracked on PRABANDH/UDISE+), fund maintenance and a permanent lab coordinator, timetable ATLs and ICT labs for Classes 6-8 project hours, and design for one projector plus shared devices	DoSEL review minutes; ASER via ThePrint
Assessment	No CT measure; out of PISA; observation journal with no template	Use Bebras-style items as a free baseline and end-of-year diagnostic, add CT items to PARAKH, and rejoin PISA for the 2029 media and AI literacy cycle	Bebras India already runs in 12 languages (Bebras India); the US discontinued NAEP TEL and now cannot measure its own progress (NAGB); AILit feeds PISA 2029 (EdTech Innovation Hub)

Two of these repairs push against CBSE's own message and need careful framing. CBSE says coding is not needed, and that is defensible for Classes 3-5 in a country where a third of schools lack a computer. It is harder to defend in Classes 6-8, where CBSE's own Middle Stage books promise "foundational knowledge on Coding" and every comparable system has pupils writing programs by age 11-12. The fix is not a device mandate. It is a **device-optional build strand**: pupils write instructions on paper, peers or the teacher "run" them, and a single projected screen shows the block version, with individual devices used only where schools have them. Language matters as well. TN SPARK's bilingual textbooks and Bebras's 12 languages show the workable pattern, while CBSE's earlier AI electives grew out of

industry partnerships such as the 2020 Intel-CBSE agreement (YourStory). Hindi or regional-language editions of the CBSE CT&AI books are not listed on the CBSE CT&AI page (CBSE CT&AI page).

6. CodeSchool should build the layer CBSE left blank

For CodeSchool, the US-India comparison sharpens one strategic conclusion and adds several design rules. The conclusion is that **CBSE has created the obligation and the timetable, and the US shows what fills that time**: coding a pupil builds, an AI model a pupil trains, a project a pupil makes, and lesson-level checks a teacher can record. CBSE supplies none of these. CodeSchool should not compete with the free books. It should sit on top of them, indexed by NCERT maths chapter, since that is the unit a CBSE teacher plans by.

CBSE gap (verified)	US reference point	CodeSchool module	What to test
Puzzles are hard, ungraded and multiple-choice (Class 3 includes 12-ball weighings)	Code.org's progressive puzzle sequences	The same chapter's skills as **levels** with warm-ups, hints and auto-checked practice, on phone, projector or paper	Do Class 3-5 maths teachers use levels in the CT minutes of a maths period?
No coding in any book; pupils never write an algorithm	Blocks from Grade 1, conditionals in code at Grade 3	**"This chapter's puzzle, now you build it"** coding levels, device-optional	Is coding a buying reason when CBSE says it is not required?
40 project hours a year in Classes 6-8, no briefs or rubrics	Project-based units in AI Discoveries; AP CSP's AI Create Project	**Project kits** (briefs, datasets, rubrics, exemplars) for 120 hours across three years	Willingness to pay; watch for CBSE's own rubric list
AI teacher books carry answer keys only	Code.org's AI teacher modules; AI teaching assistant	Teacher scripts, slides and quick checks for every AI chapter	Time saved per AI period
Hands-on AI only in Class 8, needing webcam and internet	Child trains a model in Grade 5 (CSTA); Teachable Machine in AI4K12's Grade 3-5 band	**Projector-mode ML labs** with supplied datasets for Classes 6-7, and a single-screen version of the Class 8 labs	Can one projector and one phone run a lab in a 40-minute period?
Observation journal named, no template	Lesson-level checks mapped to standards	Competency-mapped progress journal and report-card export	Do coordinators use it at term end?
No CT baseline	Bebras-style tasks	15-minute placement and end-of-year diagnostic built on Bebras-style items	Does a pre/post gain persuade principals to renew?

The US progression should shape the level design without becoming the sales pitch. CSTA 2026 gives a tested grade sequence that CodeSchool's levels can follow. Loops come before conditionals, and conditionals before variables. Machine learning appears as a yearly strand rather than one Class 8 unit. Programming emphasises reading, tracing and debugging, not just writing (CSTA 2026). Mapping every level to both CBSE competencies and CSTA codes costs little. It gives the product credibility with schools running international curricula and protects it if CBSE re-writes its own map. The AI strand is where CodeSchool can most easily get ahead of CBSE. A Class 4-5 "AI readiness" level, such as sorting objects by features or seeing how lopsided examples produce unfair results, matches US Grades 2-4. It can be done unplugged, but it should be sold as enrichment, not as a CBSE requirement. Beyond Class 8, the Class 9-10 gap before CBSE's Class 11 Python is commercially real but policy-dependent while CBSE's Class 9 plans remain unsettled.

The US also offers lessons about how products fail. Free platforms set the price of basic content at zero, and the leading ones are now entering India: Day of AI has an India programme and Experience AI publishes Indian-English units. CodeSchool's fee therefore has to come from teacher time saved, NCERT alignment, device-light delivery and assessment, not from lessons. Free platforms can also vanish, as CS First did, which supports a pitch built on continuity and local support. **Access does not equal use**: Khan Academy has said only about 15% of students with access to Khanmigo actually use it (EdTech Innovation Hub). That argues for weekly usage reporting to the school, not just licence counts. A Baltimore study found gains linked to *extra* Code.org lessons rather than the required ones (Outlier Research), which supports self-paced levels beyond the class plan. The Arkansas result is the deepest warning. If mandated provision without quality yields nothing measurable, the school that buys CodeSchool will eventually ask what changed. The diagnostic and progress record are therefore not optional features; they are what renews the contract.

Three risks follow directly from the comparison. CBSE could publish its promised project list and AI handouts at any time, so project kits must be built to wrap CBSE's topics rather than compete with them, and the CT&AI page should be checked monthly. CBSE's "no coding" message means a pitch built on "CBSE requires coding" contradicts the Board, so coding should be sold as the "build it" extension of each puzzle. And device reality caps what can be sold to government schools: the product has to work with one projector and paper before it works on tablets, because that describes the median Indian classroom.

7. Conclusion

The two systems are mirror images. The US has spent a decade building content, tools, teacher networks and devices without ever obliging most children to use them, and it now lacks even a national test to see whether any of it works. India has done the opposite in a single year: it set an obligation and generous hours for a whole board from Class 3, but supplied only puzzle banks to fill them. Each system's missing piece is the other's strength. That is the most useful framing for Indian policy, and it changes how CodeSchool should see its market. The opportunity is not "bringing American coding to India". It is supplying the middle layer that turns hours on a timetable into things children build and teachers can evidence.

The comparison also shifts which risk deserves the most attention. Before this analysis, the obvious danger was policy churn at CBSE. The Maryland-Arkansas contrast points to a quieter one: a mandate delivered without quality produces activity but no measurable result, and buyers eventually notice. A curriculum that is three years behind US standards on machine learning, and whose books never ask a child to write code, can close those gaps cheaply, on paper and on one projector, if someone builds the scaffolding. The company that measures learning, rather than the one with the most content, is best placed to keep its school customers after the first year.

8. Method, caveats and companion documents

- **Method.** Five parallel research threads (US standards and frameworks; US policy and funding; US curricula and classrooms; US access, equity and outcomes; India's wider ecosystem) wrote cited notes, which were synthesised into this report. CBSE facts follow the earlier verification note and the review of all 12 CT&AI handbooks plus the April 2026 webinar.
- **Provisional findings.** The Arkansas working paper (no detectable effect of the offer-CS mandate) was read only in fragments. The 88% US 1:1 device figure comes via a secondary repost of NCES data. Hour of AI and Code.org reach figures, and company pledges, are self-reported.
- **Gaps.** No national US figure for weekly elementary CS minutes; 2025 access data by race, rurality and income; a current count of US states with CS teacher certification; India's count of computer teachers and of teachers trained in CT/AI; CT content in Indian B.Ed. programmes.
- **Dates.** Facts are current to 6 November 2026 unless dated otherwise. Policies on both sides are changing fast (for example the US Super Intelligence Challenge relaunch in October 2026 and CBSE's promised project rubrics).

Companion documents

Document	What it adds
Research notes for this report	The five cited note files behind every section
CodeSchool Research Pack (PDF)	When programming starts in India, the level framework, market, and the full CBSE review
CBSE verification	CBSE claims checked against official circulars and syllabi
CBSE handbook review	All 12 CT&AI handbooks and the April 2026 webinar
Interview Field Kit and Research Tracker	Tools to test these conclusions with 15-20 schools